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YOLOv8-DDS: A lightweight model based on pruning and distillation for early detection of root mold in barley seedling

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ABSTRACT

Root mold proliferation presents a significant challenge in the industrial production of hydroponic barley seedlings. The small size, inconspicuous coloration, and indiscernible image of early mold regions pose new demands on detection accuracy. This study constructed a dataset of root mold in barley seedlings throughout their growth cycle and proposed the YOLOv8n-DDS detection model to integrate a lightweight detection model into a three-dimensional cyclic cultivation system. The model incorporates the dynamic sample (DySample) operator, combines deformable ConvNets v2 (DCNv2) with C2f, and reconstructs the detection head using seam carving (SEAM) technology, which enhances its capability to extract multi-scale, minute features of early-stage root mold in barley. To improve the model's performance on edge-embedded devices, this study employed layer-wise adaptive magnitude pruning and channel-wise knowledge distillation methods, thereby significantly reducing the model's parameter count and computational load. The pruned and distilled model was subsequently deployed on the Jetson Nano platform for validation. Results indicate that the YOLOv8n-DDS model outperformed the baseline model in terms of precision, recall, and mAP50 by 2.4 %, 5.6 %, and 2.2 %, respectively. The parameter count was reduced by 23.8 %, and the computational complexity (Giga floating-point operators per second) was optimized by 14.8 %. Additionally, the detection latency on resource-constrained embedded devices was further reduced by 25.8 % with TensorRT acceleration. The proposed root mold detection model is lightweight and contributes to the intelligent and technological integration of the industrial production process for high-quality barley seedling forage.

1. Introduction

Barley (*Hordeum vulgare* L.) seedlings—the tender stems and leaves harvested before the heading stage—are widely incorporated into livestock and poultry feed towing to their short growth cycle, high nutritional value, palatability, and digestibility [24]. Typically harvested at a height of 8–15 cm within 7–10 days after sowing, barley seedlings are rich in proteins, vitamins, active enzymes, and essential minerals. Hydroponic cultivation has emerged as a viable method to overcome limitations of arable land availability and environmental constraints, thereby enhancing land-use efficiency. In this system, seeds germinate and develop roots within 2–4 days, with green shoots emerging by Day 5–6. The dense root mats form a carpet-like forage layer that is suitable for direct feeding.

However, the benefits of hydroponics are offset by significant sanitary risks. The high sowing density, persistent moisture, and poor ventilation in enclosed hydroponic environments create favorable conditions for mold proliferation, which can potentially compromise the hygienic quality of the forage and reduce its economic value [1,2,29,33,34].

To address these challenges, greenhouse-based hydroponic systems have emerged as a more energy-efficient alternative to fully controlled plant factories, leveraging natural light and thermal resources. To further improve space utilization and economic efficiency, we introduce a rotatable vertical seedling cultivation rack, which allows multi-crop scheduling—supporting the cultivation of japonica rice seedlings

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during April–May and hydroponic barley in the remaining months of the year. This hardware innovation increases cultivation density and facilitates modular integration with automation systems, while also highlighting the critical need for scalable, intelligent monitoring solutions for operating under resource constraints.

Advancements in deep learning and computer vision have accelerated the digital transformation of agriculture, particularly in disease and anomaly detection [3,5,16,49]. In this context, object detection models are commonly divided into two categories: **two-stage** and **one-stage detectors**. Two-stage detectors, such as the region-based convolutional neural network (R-CNN) family, prioritize accuracy by first generating region proposals and then performing detailed classification. For instance, Wu et al. [43] developed an enhanced Faster R-CNN (TbHM-R-CNN), achieving 96.2 % accuracy in leaf disease detection—5% higher than the baseline—albeit with limited real-time applicability. Similarly, Cui et al. [12] incorporated VGG16 as the backbone of Faster R-CNN and detected weeds with 88.69 % accuracy, though at a lower inference speed compared to YOLOv7.

In contrast, **one-stage detectors, such as the You Only Look Once (YOLO) series,** have become increasingly favored for real-time applications due to their streamlined architecture and inference speed. A growing body of research have demonstrated the effectiveness of YOLO-based optimizations in agricultural scenarios. Fan et al. enhanced YOLOv5 for cotton weed detection (+9.1 % accuracy) [18], and Bist et al. achieved 98.9 % mAP@0.50 for egg cluster detection using an improved YOLO model [8]. These models have been further enhanced by targeted improvements to convolutional blocks and detection heads. For example, Xiao et al. employed deconvolution modules for real-time apple maturity grading [44], and Guan et al. [20] modified YOLOv10′s head for wheat ear counting. Moreover, innovations in feature extraction—such as those proposed by Yu et al. [48] for strawberry seedlings (92.5 % accuracy at 50.25 FPS) and Wang et al. [41] for YOLOv9′s PGI module—highlight the crucial role of architecture-level refinements when detecting **small, occluded, and morphologically variable objects**, such as mold colonies in hydroponic root mats.

Despite these advances, existing research does not specifically target mold detection in barley seedling root systems, nor addresses the dataset, model design, and deployment requirements for edge devices. To bridge this gap, we constructed the first dedicated dataset capturing the entire barley seedling growth cycle, with annotations adhering to phytopathological standards. Our objective was to build a lightweight yet high-accuracy detection model that can operate in real time under resource-limited conditions, while addressing the following limitations of existing models:

1. Microscopic and ambiguous lesion features limit conventional small-object detection.
2. Morphological polymorphism of fungal colonies exceeds the deformation tolerance of existing models.
3. High-accuracy deep models are often too large or computationally intensive for embedded deployment.

To address these challenges, leveraging the morphological diversity and multi-scale characteristics of barley root fungi, we enhanced the YOLOv8n baseline through three key modifications:(1) Integration of deformable convolutional networks v2 (DCNv2) into the C2f module; (2) Proposal of the novel dynamic sample (DySample) operator for feature sampling optimization; (3) Implementation of seam carving (SEAM) technology in detection heads.

To overcome these limitations, this study first acquired data under the latest phytopathological standards in practical barley cultivation environments, constructing the first open-source barley root mold dataset (periodically sampled throughout the complete growth cycle, which has been partially made available at https://openxlab.org.cn/datasets/wyz123/cropper. The full dataset can be obtained from the corresponding author upon request.). Then, we introduce **YOLOv8n-DDS**, a novel extension of the YOLOv8n architecture featuring:

- **DCNv2** embedded within the C2f module [14] to enhance adaptability to irregular lesion shapes;
- **DySample**, a dynamic feature sampling operator for fine-grained feature aggregation [31];
- **SEAM**, a seam-carving-based module to refine the detection head for scale-aware prediction [42].

Finally, our model achieved 97 % mAP@50 on the constructed dataset, along with a 0.5 MB reduction in model size and 1.3 giga floating-point operations per second (GFLOPs) in computational cost relative to the YOLOv8n baseline.

For practical deployment on edge hardware, we applied **layer-adaptive magnitude-based pruning (LAMP)** to reduce parameters while preserving early-stage detection sensitivity [21,25]. Additionally, we employed **channel-wise knowledge distillation (CWD)** with tailored layer matching to retain representational capacity despite pruning [22]. Deployed on a Jetson Nano with TensorRT acceleration, the final model achieved 95.1 % precision, 91.3 % recall, 94.2 % mAP@50, and a detection latency of just 57.6 ms—a 25.4 % improvement in latency compared to the unoptimized model.

In summary, this study presents **YOLO-DDS**, a lightweight and high-accuracy mold detection framework tailored for real-time deployment in hydroponic barley seedling production. Our contributions include a pioneering open-source dataset, targeted model architecture enhancements, and effective compression strategies for embedded systems. The proposed solution addresses real-world agricultural needs and supports the broader digital transformation of forage crop production.

2. Materials and methods

2.1. Dataset construction

From July to August 2024, the research team acquired data at the experimental base of the Key Laboratory of Northeast Smart Agricultural Technology, Ministry of Agriculture and Rural Affairs, Northeast Agricultural University in Harbin, Heilongjiang Province, China. The dataset construction process is illustrated in Fig. 1. During data acquisition, *Ganpi 6* barley seeds were sown in standard seedling trays (dimensions: 300 mm × 600 mm) using a precision seed drill. The trays were then placed on the rotatable vertical seedling cultivation rack. Seedling growth was monitored using Hikvision industrial cameras and laptop computers. Cameras were mounted on the horizontal movement module of the rack 0.65-meter away from the trays. RGB image sequences were generated using time-lapse photography, as shown in Fig. 1. During data acquisition, while the rack was rotating, we observed a matching relationship between the camera acquisition frequency and the rotation speed. The rack's rotation speed was dynamically adjusted based on the lighting conditions and temperature difference between the upper and lower layers, resulting in rotation variability throughout the day. We developed a camera acquisition matching program that adjusts with the rotation speed to more accurately locate the growth conditions of the barley seedlings on each tray. During the experiment, the fungal colonies on the barley roots became distinctly visible on the fourth day. Considering that the range of fungal growth expanded progressively during the growth of barley and the shading conditions varied under different light environments, data were collected from the third day to the eighth day to cover various lighting conditions. To enrich the diversity of the dataset and improve the generalization capability of the model, thereby enhancing its recognition performance in complex environments, a total of 1042 valid raw images were extracted.

This study employed LabelImg annotation software to label the image features of black and white colonies in the original images. Maximum horizontal bounding boxes were used to individually enclose each colony in the images, and the annotation format was saved as

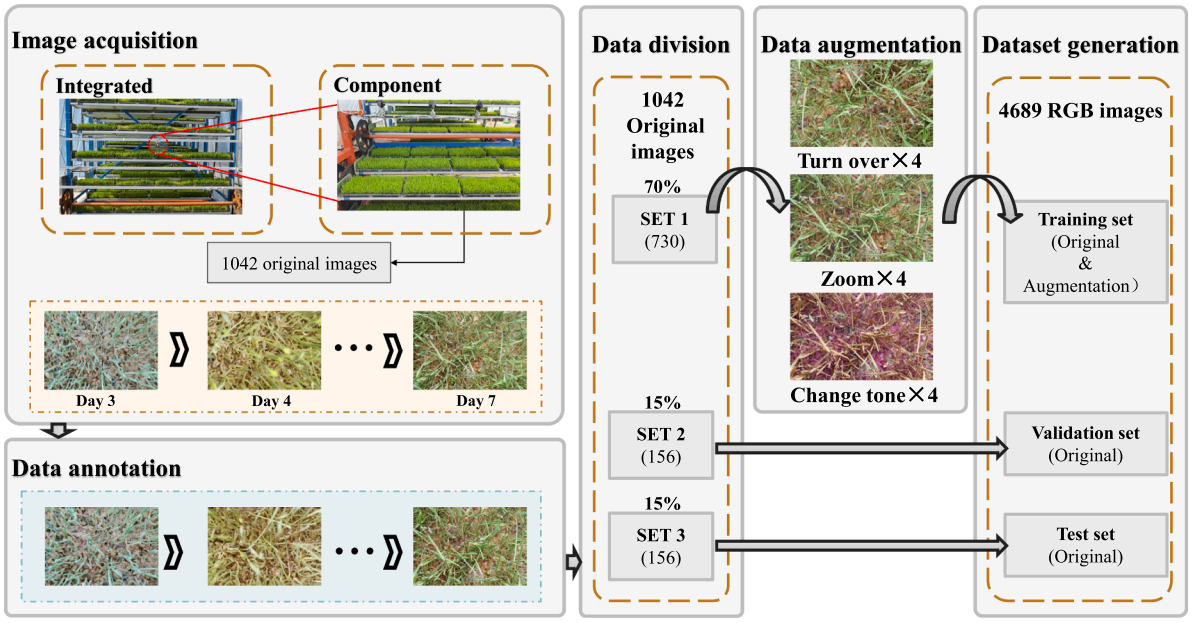


Fig. 1. Dataset construction flowchart.

YOLO-formatted txt files. The dataset was then divided into three subsets in a 70 %:15 %:15 % ratio, designated as Groups 1, 2, and 3. To prevent data leakage [7,17], data augmentation was applied to only Group 1. The augmentation techniques included turning over, zooming, tone adjustment, random flipping, addition of Gaussian noise, and hue adjustment to enhance data diversity and improve the model's generalization ability. This process generated a total of 4,380 RGB images. Finally, the augmented Group 1 data were used as the training set, and Groups 2 and 3 served as the validation and test sets, respectively.

2.2. YOLO-DDS

The YOLO series represents a typical one-stage object detection algorithm for image classification, object detection, and instance segmentation. As shown in Table 3, YOLOv8n outperforms other variants, with fewer parameters and competitive accuracy, making it ideal for mold detection tasks and efficient real-time deployment on edge devices; thus, it was selected as the baseline model.

The YOLOv8n architecture consists of four core components: input,

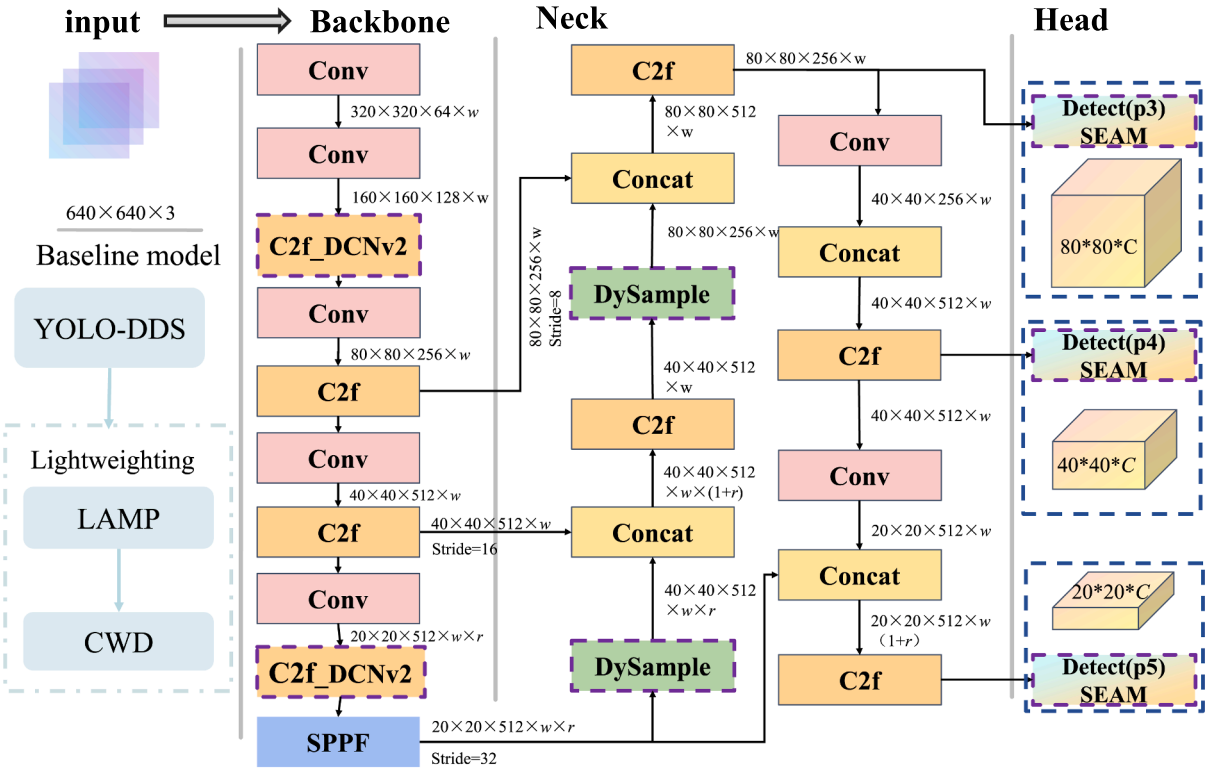


Fig. 2. YOLO-DDS architecture diagram.

backbone, neck, and head [9,35,36]. Our modifications focused on the backbone, neck, and head: (1) The backbone integrates C2f modules and spatial pyramid pooling-fast (SPPF) layers for multi-scale feature extraction [40]. (2) The neck employs an enhanced parallel attention feature pyramid network to fuse multi-scale features [30]. (3) The head combines convolutional and deconvolutional layers to generate prediction boxes, producing scale-specific outputs based on target dimensions [37].

To balance model accuracy and computational complexity while reducing model complexity and enhancing detection precision, this study developed the YOLOv8n-DDS barley root mold identification model (shown in Fig. 2). The proposed solutions are presented in subsequent sections.

2.2.1. Backbone improvement strategy

When employing the baseline model for root mold detection, fungal colonies exhibit substantial variations in morphological characteristics, coloration, and textural patterns. The structural rigidity of the C2f layer limits its adaptability to these complex variations, potentially resulting in incomplete feature extraction. Additionally, channel redundancy induced by inherent bottleneck structures within the module elevates computational overhead. To address these limitations, we strategically integrated DCNv2 with the C2f module [10,27], thereby enhancing the model's adaptability and operational flexibility.

During convolutional operations, DCNv2 introduces learnable offset parameters $(\Delta x, \Delta y)$ generated by lightweight convolutional layers. This mechanism predicts sampling point displacements and adjusts feature map sampling accordingly, thereby effectively capturing object boundaries and morphological features. The mathematical formulation is detailed in Eq. (1), with structural configuration illustrated in Fig. 3.

$$y(i,j) = \sum_m \sum_n x(i + \Delta_{x_m}, j + \Delta_{y_n}) \bullet k(m,n) \tag{1}$$

where x denotes the input feature map, k represents the convolution kernel, and Δ_{x_m} corresponds to the learned offsets, and (i,j) indicates coordinates in the output feature map. This mechanism enables DCNv2 to dynamically adapt to target shape variations, thereby improving detection accuracy.

This study replaced the convolutional layer in the final feature layer of the backbone with DCNv2, enabling the convolutional kernels to adaptively adjust to shape variations in fungal colonies. This modification aligns sampling operations with the actual shapes and sizes of colonies, effectively improving detection accuracy and model robustness while reducing computational complexity.

2.2.2. Neck improvement strategy

To enhance multi-scale feature extraction capability, this study incorporated DySample into the head network. For small mold regions, the module allocated larger sampling factors to improve the extraction of fine features, while adaptively adjusting sampling operators for larger infected areas to prevent feature distortion. By learning sampling points directly, the approach eliminated time-consuming dynamic convolution operations and auxiliary subnetworks required in conventional kernel-based dynamic upsamplers, significantly reducing computational overhead [28,45]. The architectural configuration is depicted in Fig. 4.

In the preliminary design of DySample, the sampling positions for each point in the input feature map X were initially fixed at identical locations(standard grid points). This initialization scheme is termed "nearest initialization". To improve this mechanism, DySample adopted "bilinear initialization", where zero offsets yielded bilinearly interpolated feature maps. The module generated sampling offsets O through a linear layer, which were subsequently applied to resample continuous feature maps. This process can be mathematically expressed as follows:

$$O = LinearLayer(X) \tag{2}$$

where O denotes the generated offsets and X represents the input feature map.

The process utilized PyTorch's grid sampling function to resample the input feature map X based on the generated offsets O and original grid positions G, producing the upsampled feature map X'. This operation can be mathematically formulated as follows:

$$X' = GridSample(X, G + O) \tag{3}$$

where G denotes the original grid positions and O represents the generated offsets.

DySample employed bilinear interpolation to pre-interpolate the input feature map, generating continuous feature representations. Dynamic sampling points were then applied to the interpolated features for resampling. The bilinear interpolation process is mathematically defined as follows:

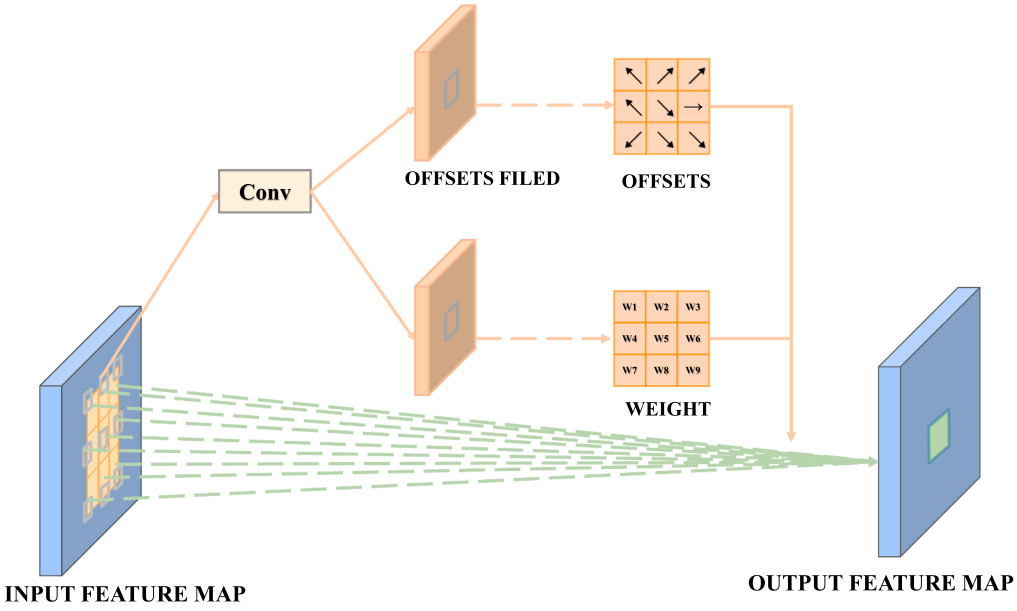


Fig. 3. DCNv2 architecture.

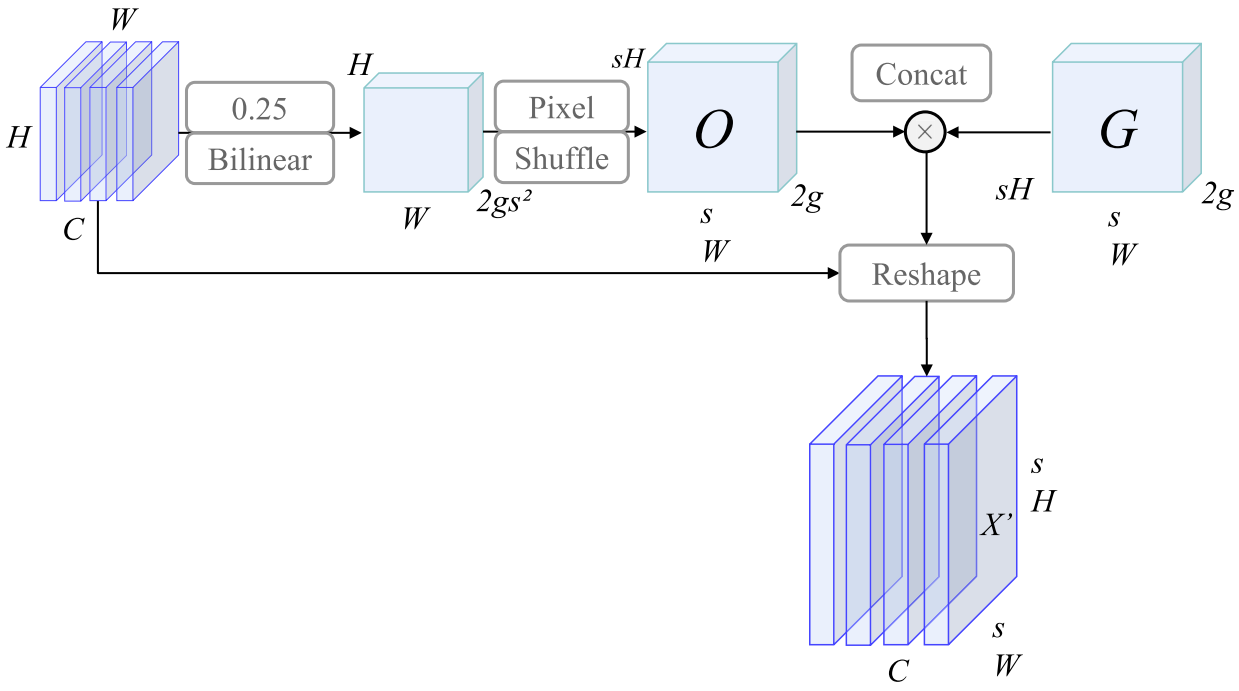


Fig. 4. DySample architecture diagram.

$$I(x,y) = (1-u)(1-v)I(x-1,y-1)+ u(1-v)I(x+1,y-1)+ (1-u)vI(x-1,y+1)+ uvI(x+1,y+1)\# \quad (4)$$

where $I(x,y)$ denotes the interpolated pixel value at position (x,y), and u and v represent the interpolation ratios along the x and y axes, respectively.

DySample optimized the sampling process to reduce parameter quantity, computational load (floating point operations per second), Graphics Processing Unit memory consumption, and latency. Compared with kernel-based dynamic upsamplers, DySample requires no additional CUDA packages, while achieving fewer parameters and lower computational costs, along with superior performance across various dense prediction tasks.

2.2.3. Head improvement strategy

Traditional detection heads suffer from excessive computational resource consumption and parameter redundancy, which limits their deployment in practical agricultural applications. To enhance model performance on edge devices, this study integrates the content-aware image scaling technique, SEAM, into the detection head. SEAM combines scale-equilibrium attention mechanisms with content-aware image scaling strategies for efficient data mining [4]. The scale-equilibrium attention mechanism evaluates the relative importance of multi-scale features in detection tasks and dynamically adjusts computational resource allocation, thereby effectively optimizing resource distribution. The content-aware scaling strategy adaptively modifies image dimensions by selectively removing or inserting pixel seams, while preserving critical image content and foreground objects. This approach identifies optimal pixel paths based on image energy functions, iteratively eliminating low-importance seams or inserting new paths in less significant regions, thereby maintaining visual content

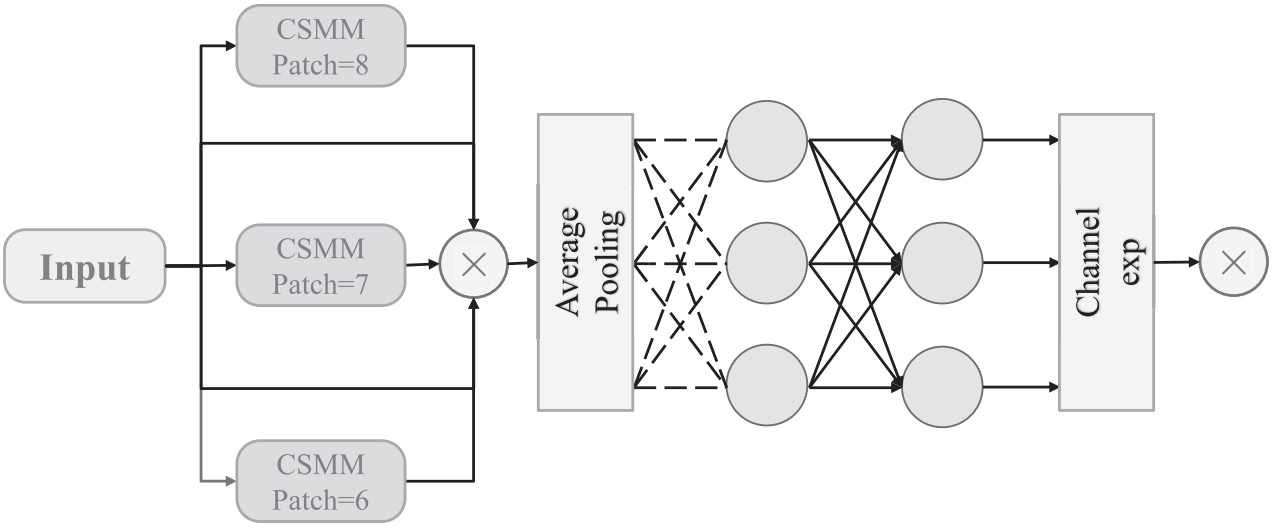


Fig. 5. SEAM-Head architecture diagram.

integrity more effectively than conventional scaling methods.

Fig. 5 illustrates the schematic diagram of the SEAM-Head architecture. The input image initially passes through three consecutive continuous seam mapping modules (CSMMs). These modules perform feature extraction and representation enhancement through cascaded convolutional layers, activation functions, and potential normalization layers. Following spatial dimension reduction via an average pooling layer that preserves critical feature information, the head module executes final object detection tasks.

The complete workflow can be mathematically represented by the following formula:

$$OutPut = SEAMHead(AveragePooling(CSMM_3(CSMM_2(CSMM_1(Input))))) \quad (5)$$

2.3. Model lightweighting

Although YOLOv8n-DDS demonstrates high detection speed and accuracy in field experiments, effectively accomplishing barley root mold colony detection tasks, its substantial parameter count and redundant architectural components limit practical deployment. To reduce model complexity and streamline the architecture for deployment on resource-constrained edge computing devices, this study implemented LAMP for structural compression.

We employed a tensor-based LAMP scoring mechanism that evaluates channel importance through L2-norm analysis. The LAMP score was computed by normalizing the squared magnitude of each weight against the sum of squared magnitudes of all remaining weights within the same layer, where lower scores indicate weaker weight significance, prioritizing these connections for elimination by pruning, expressed as [25,47]:

$$score(u; W) = \frac{(W(u))^2}{\sum_{v \geq u}(W(v))^2} \quad (6)$$

The numerator corresponds to the squared magnitude of the target connection weight, and the denominator represents the sum of squared magnitudes from all remaining connections within the same layer. This formulation enables LAMP scores to effectively quantify the relative importance of connections within their respective layers. By applying a user-defined global pruning ratio, the method automatically determines layer-specific sparsity levels that satisfy global sparsity constraints while guaranteeing at least one optimal channel (with unit score) per layer, thereby mitigating the layer collapse risk inherent in conventional global pruning approaches [50]. The LAMP process can be described as Fig. 6.

Knowledge distillation enables performance migration in neural networks. Specifically, this involves transferring capabilities from a high-performing but complex and bulky teacher model to a more compact and computationally efficient student model. This technique allows the student network to retain the majority of the teacher's performance while significantly reducing memory footprint and computational costs. This study also employed CWD, a feature alignment methodology operating at the channel granularity. By minimizing distributional discrepancies between channel-wise features of teacher and student models, the approach achieved fine-grained knowledge transfer. The absence of complex feature transformations renders this method particularly suitable for lightweight models operating under computational constraints. CWD is a knowledge distillation method specifically focused on the channel dimension [39]. By normalizing the activation values of each channel, they are converted into probability maps. Further, channel-level probability discrepancies between student and teacher networks are minimized using asymmetric Kullback-Leibler (KL) divergence [46]. Thus, CWD emphasizes learning significant regions within individual channels, thereby improving performance on dense prediction tasks. Knowledge distillation schematic diagram is shown as Fig. 7.

CWD proposes channel-wise normalization to convert activation maps of each channel into probability distributions, where the activation values within each channel can be treated as probability maps. Through KL divergence minimization, the method reduces channel-level probability map discrepancies between teacher and student networks, enabling the student network to better learn the most salient regions in the teacher network. This enables the student network to capture the most discriminative regions within each channel of the teacher's outputs.

The specific operational procedure involves normalizing the activation values of each channel in the teacher network into probability distributions using the SoftMax function, calculated as follows:

$$\phi(y_c) = \frac{exp\left(\frac{y_{c,i}}{T}\right)}{\sum_{i=1}^{W \bullet H}\left(\frac{y_{c,i}}{T}\right)} \quad (7)$$

where $y_{c,i}$ denotes the activation value at the c position of the i channel, and T represents the temperature coefficient in distillation.

To better facilitate the student model's learning, CWD employs the KL divergence between channel-level probability maps of teacher and student networks as the loss function. During the distillation process, the student network's learning is guided by minimizing this divergence,

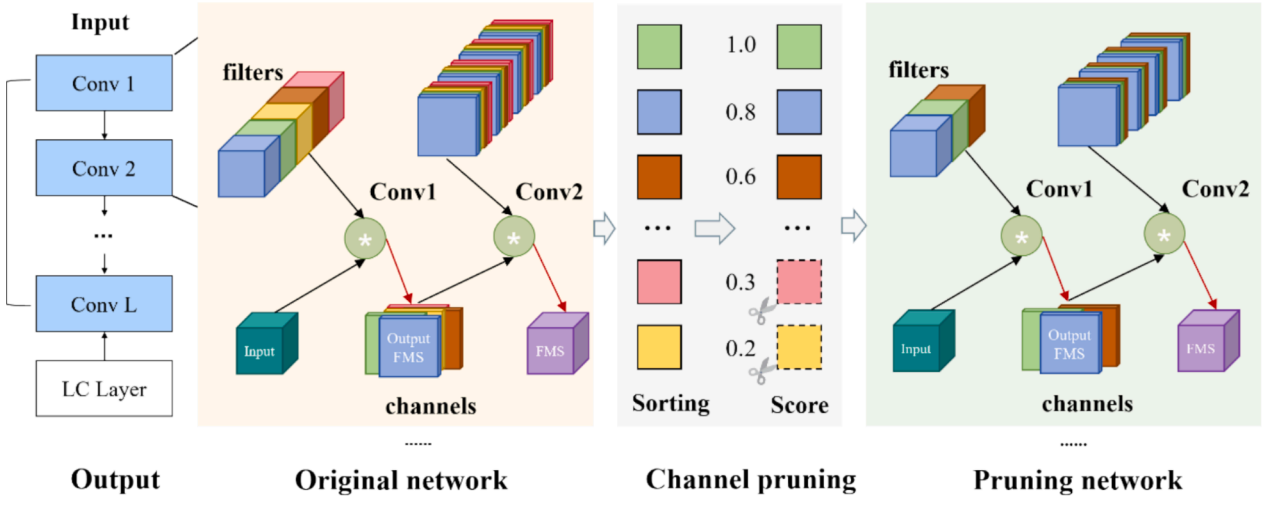


Fig. 6. Diagram of the LAMP process.

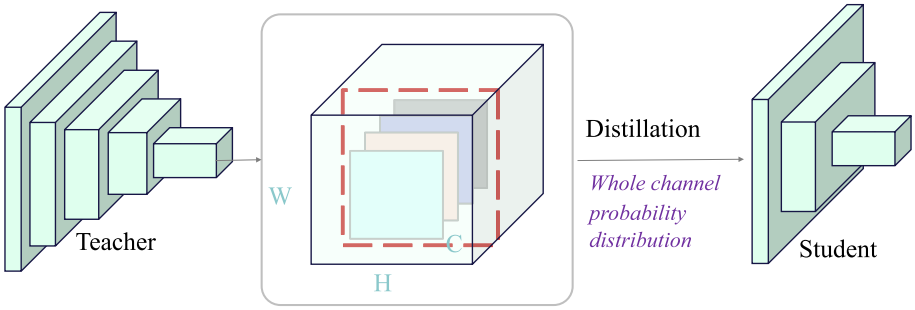


Fig. 7. Knowledge distillation schematic diagram.

which is formulated as follows:

$$\varphi\left(y^T, y^S\right) = \frac{T^2}{C}\sum_{c=1}^{C}\sum_{i=1}^{W*H}\phi\left(y_{c,i}^T\right) \bullet \log\left(\frac{\phi\left(y_{c,i}^T\right)}{\phi\left(y_{c,i}^S\right)}\right) \tag{8}$$

2.4. Experimental Setup

All experiments were conducted under identical conditions to ensure unbiased results. The trials were performed on a 64-bit Windows operating system. The detailed server configurations and the hyperparameters used in the experiment are provided in Tables 1 and 2, respectively.

2.5. Evaluation metrics

Three primary metrics were employed to assess the model accuracy: precision (P), recall (R), and mean average precision (mAP). The computational formulations are as follows:

$$P = \frac{TP}{TP + FP} \tag{9}$$

$$R = \frac{TP}{TP + FN} \tag{10}$$

$$mAP = \frac{1}{n}\int_1^n AP_i \tag{11}$$

Four key metrics were used to evaluate model lightweighting in detection: parameter count (Parameters), GFLOPs, model size (Weight), and inference latency per input sample (Latency). The computational metrics can be defined as follows:

$$Parameters = \sum_{i=1}^{n} W_i + b_i \tag{12}$$

$$GFLOPS = \frac{2 \times M \times N \times K}{1024^3 \times timeInSec} \tag{13}$$

$$Detection\boldsymbol{Latency} = \frac{1000}{\boldsymbol{Fps}} \tag{14}$$

Table 1
Server configuration.

mirror	PyTorch 1.11.0
	Python 3.8(ubuntu20.04)
	Cuda 11.3
GPU	RTX 2080 Ti(11 GB)*1
CPU	12 vCPU Intel(R) Xeon(R) Platinum 8255C CPU @ 2.50 GHz

Table 2
Hyperparameter settings during training.

Parameter	Value
Epochs	200
Learning rate	0.0001
Batch size	32
Workers	8
Image size	640
Optimizer	SGD
Test IoU	0.5
Weight decay coefficient	0.0005

3. Results

3.1. Comparative experiments on detection models

To determine the most appropriate baseline model, a comparative analysis was conducted on lightweight detection baseline models, including YOLOv5n, YOLOv5s, YOLOv8n, YOLOv8s, YOLOv10n, YOLOv10s, YOLOv11n, YOLOv11s, YOLOv12n, YOLOv12s, RT-DETR-L, and RT-DETR-X, all characterized by few parameters.

As shown in Table 3, within the n-series models, YOLOv8n achieved the highest mAP50 and precision values, 96.3 % and 96.5 %, respectively, with its weights being only 0.1 higher than that of YOLOv12n, while maintaining a lower computational cost of 8.1 GFLOPs compared to YOLOv10n. Compared to s-series models, YOLOv8n exhibited superior mAP50 and precision over YOLOv5s and YOLOv11s, approaching the performance levels of YOLOv8s, YOLOv10s, and YOLOv12s, while retaining extremely lightweight characteristics. Based on the comprehensive evaluation of detection performance and computational efficiency, YOLOv8n was selected as the baseline model for this study. The radar chart in Fig. 8 visually compares the multi-dimensional performance metrics across all evaluated models.

Additionally, RT-DETR was included in the comparative analysis. However, although it demonstrated high detection accuracy, its large parameter size and computational demands limited its suitability for embedded, real-time applications in this study.

3.2. Ablation experiments

Ablation experiments were conducted to validate the effectiveness of the DCNv2 module, DySample module, and SEAM detection head in barley root mold identification. As shown in the experimental results presented in Table 4, the incorporation of the DySample module in Experiment 3 resulted in higher precision, recall, and mAP, compared to the YOLOv8n baseline model, while maintaining comparable parameter size and GFLOPs, demonstrating the DySample module's effectiveness in enhancing model recognition performance.

The configuration in Experiment 4, where the SEAMHead module was added solely to the baseline model, achieved limited improvement in detection performance, but exhibited significant reductions in

Table 3
Comparison of detection results across baseline models.

model	Precision P/%	Recall R/%	mAP50/%	Weights /MB	parameters	GFLOPs
YOLOv5n	86.1	80.9	87.1	3.8	1,761,871	4.1
YOLOv5s	90.3	90.1	90.1	14.3	7,015,519	15.8
YOLOv8n	96.5	91.0	96.3	6.2	3,006,038	8.1
YOLOv8s	97.1	92.3	97.0	11.2	11,405,876	28.4
YOLOv10n	95.7	93.4	95.5	7.1	3,723,459	9.7
YOLOv10s	96.3	93.1	97.1	16.5	8,036,508	24.4
YOLOv11n	94.4	88.3	91.3	6.3	2,582,542	6.3
YOLOv11s	96.1	93.4	94.5	21.4	14,872,566	36.8
YOLOv12n	96.3	94.7	96.0	6.1	3,108,742	7.9
YOLOv12s	97.1	93.4	96.4	19.7	11,905,334	31.6
RT-DETR-L	96.5	93.0	95.5	20.0	29,428,460	105.5
RT-DETR-X	97.0	93.5	96.2	35.0	67,467,852	232.7

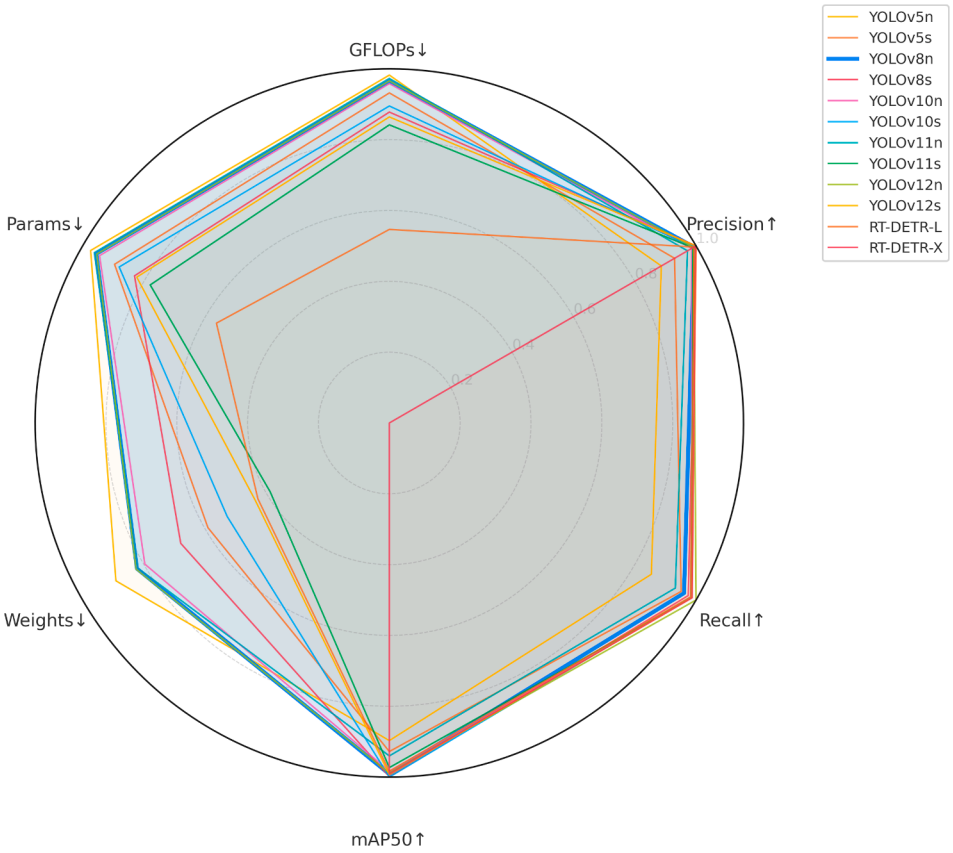


Fig. 8. Radar chart of detection performance and model complexity across baseline models. Note: Detection performance metrics (mAP50, Recall, Precision) shown in this figure were normalized to their maximum values, whereas the lightweight indicators (Weights, Parameters, GFLOPs) were scaled to reflect model efficiency.

Table 4
Ablation study results.

Test No.	DCNV2	DySample	SEAMHead	Precision P/%	Recall R/%	mAP50/%	mAP50-95/%	Weights/MB	Parameters	GFLOPs
1	×	×	×	0.965	0.91	0.963	0.732	6.2	3006038	8.1
2	√	×	×	0.949	0.875	0.94	0.673	6.3	3037425	8.0
3	×	√	×	0.966	0.926	0.969	0.753	6.3	3018390	8.1
4	×	×	√	0.942	0.855	0.929	0.653	5.9	2817878	7.0
5	√	√	×	0.952	0.885	0.947	0.692	6.3	3049777	8.0
6	×	√	√	0.971	0.92	0.967	0.746	5.9	2830230	7.0
7	√	×	√	0.951	0.877	0.943	0.674	6.0	2849265	6.9
8	√	√	√	0.969	0.932	0.97	0.755	5.7	2861617	6.9

parameter size, number of parameters, and FLOPs, demonstrating the SEAMHead module's superior lightweighting capabilities.

Experiment 6, which incorporated both the SEAMHead and DySample modules into the baseline model, demonstrated increased precision, recall, and mAP (mAP@0.5 and mAP@0.5:0.95) showing significant improvements over the baseline. This indicates that the

integration of SEAMHead and DySample modules substantially enhances detection performance. Furthermore, the model exhibited reduced parameters (2,830,230 vs. 3,012,450 in Exp.3), lower weight size (5.9 MB vs. 6.5 MB), and decreased computational complexity (7.0 GFLOPs vs. 8.1 GFLOPs), demonstrating effective integration of both modules to ensure improved detection accuracy while achieving lightweight performance through reduced model size and computational demands.

Experiment 7, integrating both DCNv2 and SEAMHead modules, demonstrated that although the standalone DCNv2 integration (Experiment 2) underperformed compared to the YOLOv8 baseline, the combined configuration showed improved precision and recall relative to Experiment 4 (SEAMHead-only), indicating synergistic performance enhancement through module combination.

Experiment 8(YOLOv8-DDS) integrating all three modules demonstrated a 2.2 percentage point improvement in recall, the lowest weight size (5.7 MB), and 1.1 s reduction in FLOPs, with substantial gains in precision and mAP. Compared to Experiment 5 (DCNv2 + DySample), the SEAMHead-enhanced configuration showed significant improvements across all detection accuracy metrics and model compactness, indicating that the tri-module combination maximally optimized the YOLOv8n baseline to achieve optimal comprehensive performance.

3.3. Feature visualization

To elucidate how the model enhances recognition of fine-grained features, this study visualized fused feature maps to reveal its feature extraction capabilities and decision-making rationale. Each model layer contained multiple channel feature maps; therefore, layer-wise feature fusion was implemented using the LAMP scores to assess channel importance. The fusion weights were computed using these scores, as detailed in Eq. (15):

$$F_{fused} = \sum_{c=1}^{c} \omega_c \bullet F_c \tag{15}$$

$$\omega_c = \frac{LAMP_score_c}{\sum_{k=1}^{C} LAMP_score_c} \tag{16}$$

The comprehensive feature map analysis is presented in Fig. 9. As illustrated, comparative analysis of the final convolutional layer feature maps between the YOLOv8-DDS and YOLOv8n backbones under varying illumination conditions (8:00 AM, 12:00 PM, and 4:00 PM) revealed significantly stronger activation responses in disease-affected regions within YOLOv8-DDS visualizations. These observations indicate that the architectural modifications to the YOLOv8n backbone substantially enhanced feature extraction capability across all tested environmental lighting variations.

Heatmaps are color-gradient representations in which regional luminance intensity corresponds to the model's attention weights in prediction outcomes, with brighter areas indicating higher attentional focus during detection. Fig. 10 presents heatmap visualizations across different models during barley root mold development, revealing that YOLOv8-DDS exhibited concentrated heatmap activation along hyphal edges of mold colonization (red-highlighted areas), with intensified chromaticity indicating higher activation intensity, and a significantly larger coverage area than other models. In contrast, YOLOv8n and YOLOv9 displayed dispersed activation patterns with partial response shift toward root system backgrounds (blue-tinted regions), reflecting weaker target-specific perception. Although YOLOv11 adequately covered lesion core zones, it demonstrated an attenuated response to hyphal expansion margins (yellow-arrow-indicated areas).

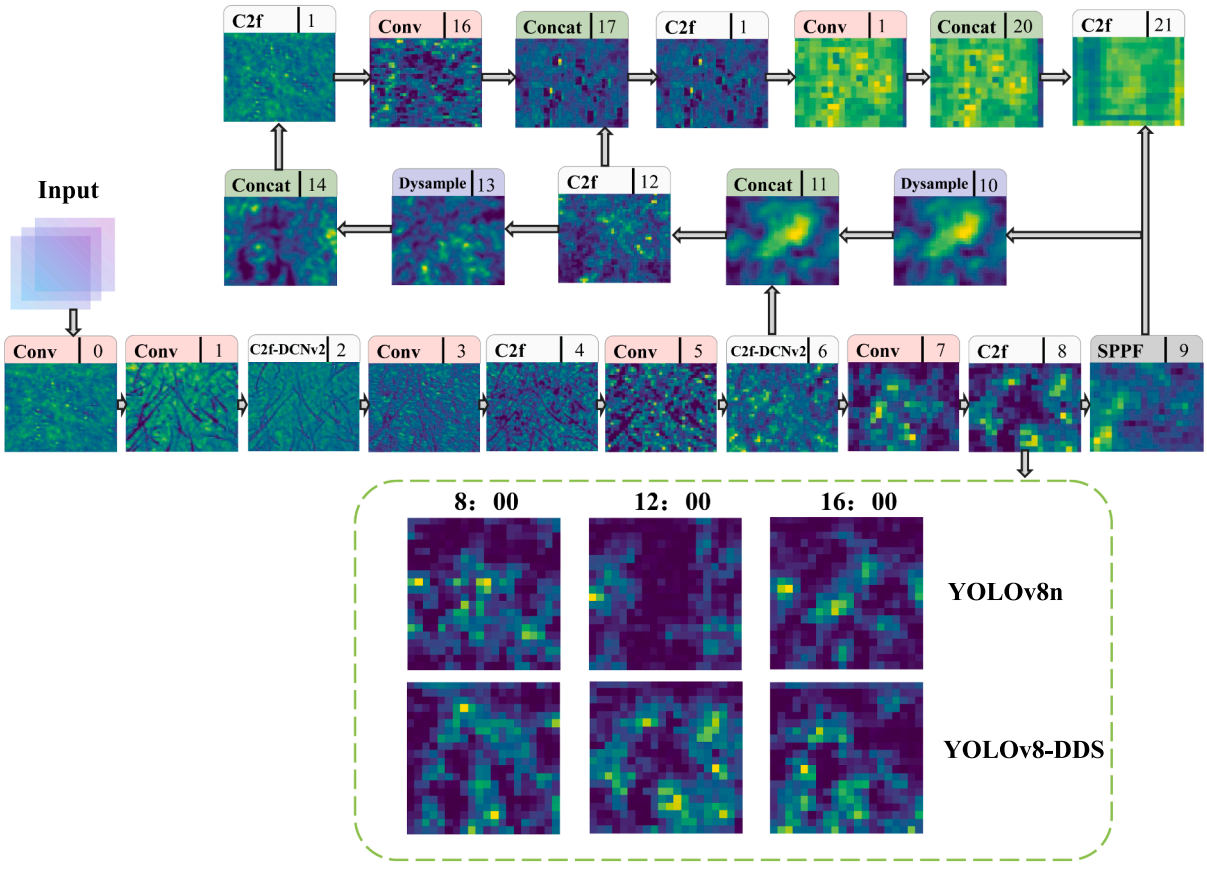


Fig. 9. Feature maps across model layers.

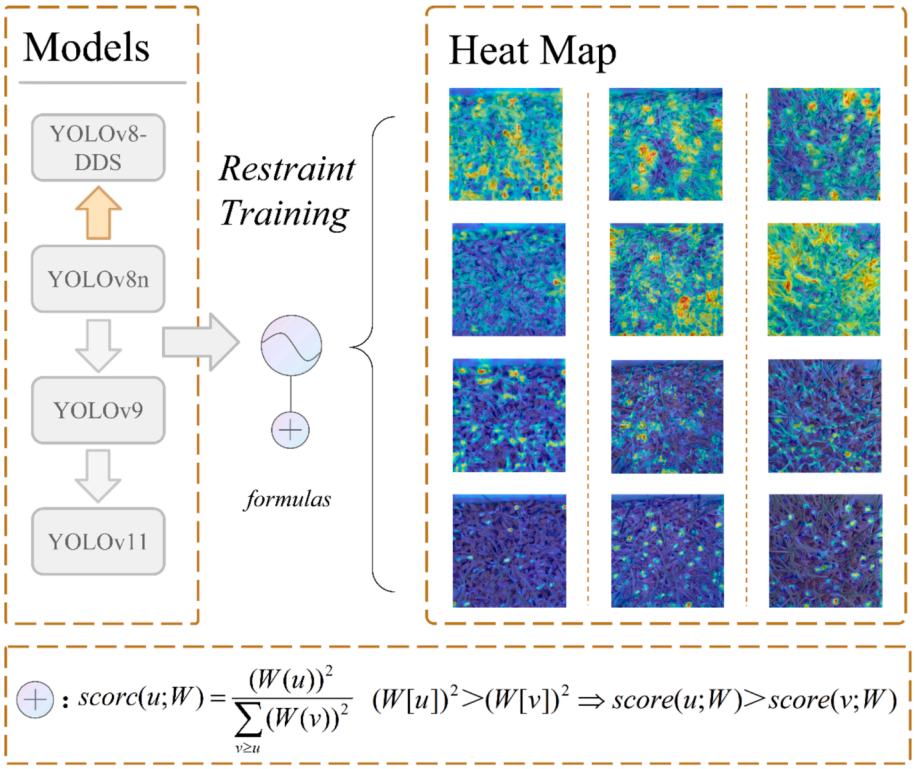


Fig. 10. Heatmap visualizations across different models during barley root mold development.

3.4. Lightweight deployment

This study implemented LAMP for model compression, achieving an 80.09 % pruning ratio (parameter count ratio) that eliminated approximately 20 % redundant parameters while maintaining critical weights. The pruned model achieved a significant performance improvement. The layer-wise pruning effects are visually demonstrated in Fig. 11.

In this study, the trained YOLOv8-DDS was designated as the teacher network, while the pruned YOLOv8-DDS served as the student model. Two distillation strategies were implemented: CWDLoss and multi-granularity distillation (MGD) loss. The comparative performance of the distilled student models is summarized in Table 5, highlighting the effectiveness of knowledge transfer between teacher and student architectures.

Both methods maintained comparable levels of recall (%) and mAP@0.5 (%), whereas CWD outperformed MGD by 0.9 percentage points in terms of precision (%) and 0.4 pp in mAP@0.5:0.95. This phenomenon is attributed to layer-independent distillation employed by

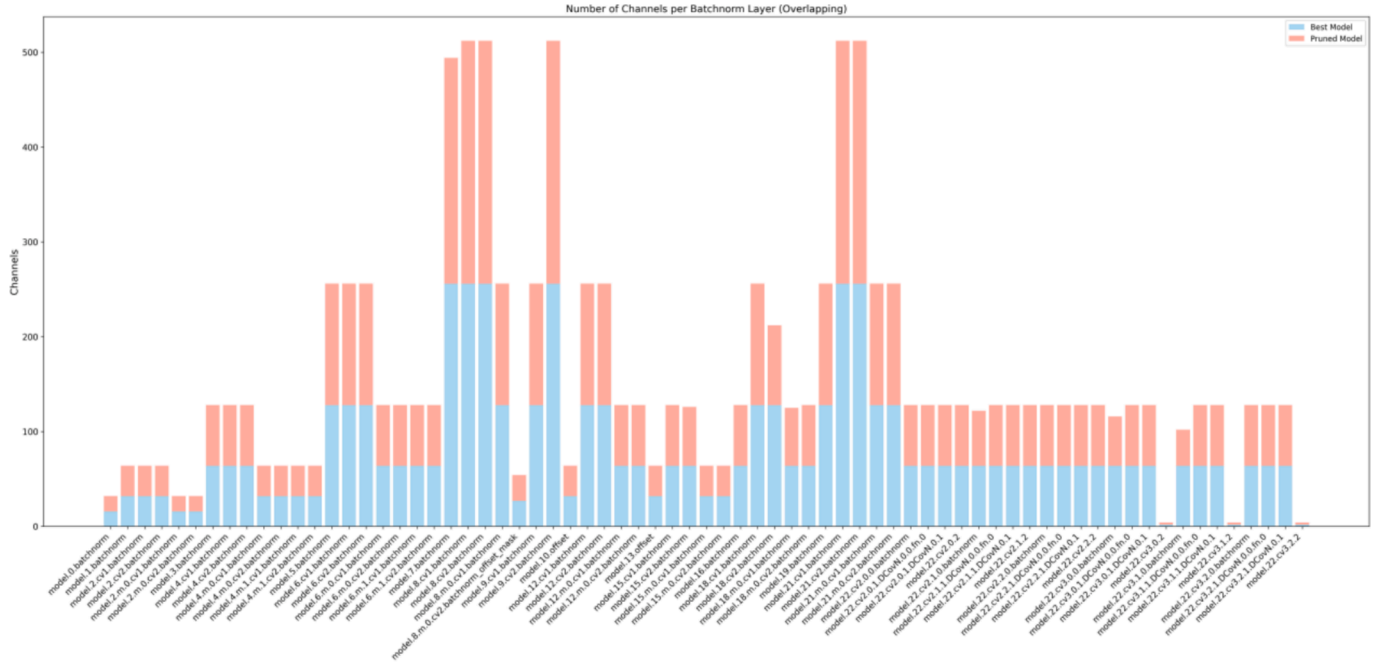


Fig. 11. Pruning process and redundancy removal.

Table 5
Distillation performance comparison.

Methods	Precision/ %	Recall/ %	mAP50/ %	mAP50-95/ %	Weights/ MB
MGD	98	96.6	985	83.3	5.7
CWD	98.9	96.6	98.5	83.7	4.6

CWD across multi-level channels, thereby adaptively adjusting feature distributions for multi-scale targets. Such adaptability proves critical for capturing microscale variations in barley root mold morphology, making CWD particularly suitable for datasets with small targets and scale-varying characteristics.

After pruning and distillation, the lightweight model was deployed on a Jetson Nano-based physical machine to validate operational stability. Utilizing the TensorRT inference framework, the deployment process involved exporting the trained model weights (best.pt) to an ONNX-formatted intermediate file. This was followed by a Transpose node being inserted before the output layer via the v8trans.py script in the infer directory to generate best.transd.onnx, and the trtexec tool was finally used to compile the TensorRT engine. This approach enabled high-performance inference with architectural decoupling, eliminating complex encapsulation.

To validate the operational performance of the models in practical applications, this study conducted comparative tests on the Jetson Nano platform, evaluating the original optimized model against its pruned and distilled counterpart, and the performances are shown in Table 6.

4. Discussion

This study addresses three critical challenges in barley root mold detection tasks: (1) insufficient detection accuracy due to extremely small target features, (2) inadequate deformation robustness caused by morphological variations in target structures, and (3) the inherent trade-off between computational efficiency and detection accuracy [13]. To overcome these limitations, we propose a dedicated detection model optimized for barley root mold identification.

4.1. Justification for Selecting YOLOv8n as the baseline model

Although the YOLO series models share a broadly similar architecture, key differences in module design across versions significantly impact their performance on small target detection tasks. The structural innovations in YOLOv8n, particularly the replacement of the CSP block with the C2f module, enhance gradient flow and reduce parameter redundancy, effectively preserving micro-target features in complex backgrounds. The SPPF module efficiently fuses multi-scale contextual information at reduced computational costs, while also improving the model's capability to detect targets at different scales. The decoupled head design further separates the learning paths for classification and regression tasks, enhancing adaptability in small object detection.

By contrast, the SDDS module introduced in YOLOv10n aims to enhance feature utilization and limit information loss, which benefits global perception but has limited ability to preserve edge details critical for small target detection. The C3k2 and C2PSA modules in YOLOv11n improve computational efficiency through group convolution and lightweight bottleneck structures, whereas the improvement in small target detection tasks remains limited. YOLOv12n integrates the BiFormer module to enhance global modeling capacity, which is beneficial for large targets and dense scenes. However, it has limited suitability for early-stage, small lesion detection, and it substantially increases parameter counts and computational overhead.

Therefore, the architectural design of YOLOv8n strikes a better balance between lightweight deployment, feature extraction capability, and adaptability for embedded scenarios, which makes it a reasonable baseline choice for this study. The comparative analysis and experimental results further validate the superior suitability of YOLOv8n for small object detection tasks such as early-stage barley root mold identification.

It is worth noting that although transformer-based models like RT-DETR demonstrate competitive detection accuracy, their high computational complexity and resource demands make them less suitable for real-time, embedded deployment scenarios compared to YOLOv8n.

4.2. Architecture improvements for small target detection

The high complexity of detection targets presents optimization opportunities in YOLOv8n's architecture for barley root detection: (1) In the backbone, the rigid structure of the original C2f modules failed to adapt to complex root colonization patterns, leading to incomplete feature extraction, while internal bottleneck structures caused channel redundancy. We replaced partial C2f modules with DCNv2 to enhance adaptability [23]. (2) In the neck, the fixed bilinear interpolation strategy inadequately preserved multi-scale feature details. DySample was implemented to dynamically adjust sampling positions based on mold colony size, compensating for information loss during upsampling [32]. (3) In the head, the decoupled head structure involved excessive computations on irrelevant background channels. The SEAM channel attention mechanism was adopted to prioritize critical features [6]. These three improvements collectively addressed deformation adaptability, scale sensitivity, and computational efficiency, as demonstrated in Table 4.

4.3. Pruning and knowledge distillation for lightweight deployment

Although YOLOv8-DDS achieves an optimal balance between accuracy and lightweight design, its deployment on agricultural edge devices remains constrained by computational resource limitations and stringent real-time requirements [26]. To overcome these limitations, this study implemented a dual optimization strategy combining pruning-quantization co-optimization and knowledge distillation.

Current mainstream pruning methodologies primarily include unstructured and structured pruning [25]. Unstructured pruning removes near-zero weights for model lightweighting but risks over-pruning low-magnitude parameters in shallow layers, which are critical for detecting microscale features in barley root mold. Thus, we adopted structured pruning. Conventional structured approaches, such as BN channel pruning and layer pruning, employ static criteria and fail to optimize pruning configurations for target-specific features [38]. This study implemented LAMP pruning [19], which dynamically allocates pruning rates across layers based on their functional characteristics. By prioritizing retention of critical shallow-layer feature extraction capacity tailored to microscopic mold morphology, LAMP effectively resolved the

Table 6
Accelerated deployment model performance comparison.

Models	Weights /MB	Precision /%	Recall /%	mAP50 /%	Detection Latency /ms	TensorRT Acceleration
YOLOv8n	5.9	95.2	95.7	97.7	150.7	×
	27.2	94.7	95.5	97.7	77.2	√
YOLOv8n-LC	4.0	95.8	92.2	95.2	112.4	×
	16.1	95.1	91.3	94.2	57.6	√

static evaluation limitations inherent in traditional structured pruning.

This study employed knowledge distillation synergistically with pruning to restore detection performance while reducing model size [11]. Mainstream distillation approaches include response-based, relation-based, and feature-based knowledge transfer [15]. Response-based methods directly align output layers but neglect intermediate feature propagation that is critical for detecting microscale root mold targets. Relation-based distillation constrains inter-feature distances; however, unstable morphological variations in mold colonies hinder reliable relation modeling. Thus, feature-based distillation was prioritized, with CWD and MGD selected as representative methods. CWD transfers channel-wise features, yielding compact models ideal for edge devices, whereas MGD emphasizes pixel-level spatial correlations. As demonstrated in Table 5, CWD's channel-focused approach achieved superior deployment efficiency on resource-constrained hardware.

5. Conclusion

This study proposes an efficient method for detecting barley root mold, providing technical support for accurate and high-efficiency detection. The research first selected YOLOv8n as the baseline model due to its balanced precision and lightweight performance. Subsequently, targeted improvements were made based on the characteristics of root mold, followed by further lightweight optimization through LAMP and CWD knowledge distillation. The final model achieved a precision of 98.9 %, recall of 96.6 %, and mAP@0.5 of 98.5 %, representing improvements of 2.4 %, 5.6 %, and 2.2 % over the original model, respectively. Additionally, the model reduced parameters by 23.8 % and improved GFLOPS by 14.8 %. On the Jetson Nano platform, the detection latency reduced from 112.4 ms to 57.6 ms after acceleration, corresponding to 25.4 % and 25.3 % improvements over the original model, meeting real-time detection requirements.

Overall, lightweight deployment reduced the hardware and computational costs, thereby achieving high-precision detection of root mold proliferation. This ensures timely interventions to suppress mold growth using biological, chemical, or environmental control methods. Consequently, this method enhances the intelligent unmanned management of industrialized hydroponic barley seedlings production and offers technological support for reducing the costs of livestock breeding and improving forage quality.

6. Data availability

The dataset supporting the findings of this study has been partially released on OpenXLab at the following link: https://openxlab.org.cn/datasets/wyz123/cropper. The complete dataset is available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Huang Junjie: Methodology, Writing – original draft, Validation, Data curation, Visualization. **Ma Zheng:** Writing – review & editing, Data curation, Methodology. **Wu Yuzhu:** Data curation, Methodology. **Bao Yujian:** Data curation, Investigation. **Wang Yizhe:** Data curation, Visualization. **Su Zhongbin:** Resources, Supervision, Funding acquisition. **Guo Lifeng:** Writing – review & editing, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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